

SHAVEJ SHAIKH

Mumbai | +91 7738288136 | shavejshaikh1998@gmail.com | <https://www.linkedin.com/in/shavejshaikh> | <https://shavejshaikh.github.io>

TECHNICAL SKILLS

Languages: Python, SQL, C

Backend & Frameworks: FastAPI, Django, Flask

Cloud & DevOps: AWS(S3, EC2, Lambda, Cloudwatch, Athena), Docker, Airflow

Data & BigData: Kafka, Elasticsearch, Polars, Pandas

Database: PostgreSQL, MySQL, MongoDB

Framework & Libraries: FastAPI, Django, Flask, Polars, Redis, RabbitMQ, Pandas, Docker

AL / ML: Tensorflow, OCR(Tesseract, PaddleOCR, EasyOCR), YOLO

Generative AI & LLMs: LangChain, LlamaIndex, LangChain, Mem0, PgVector, RAG, Prompt Engineering

Tools: Git, Bitbucket, Gitlab, Github, Airflow, Linux, Cursor

PROFESSIONAL EXPERIENCE

Nanoprecise Sci Corp | Bangalore, India

October 2023 - Present

Software Engineer

- **Revamped DTM and CDP legacy systems**, reducing data processing time from **28 hours to 3 hours** for 16K+ tag IDs using **multi-processing, optimized I/O, and NumPy/Pandas**, while improving CDP efficiency by **40%** through algorithmic and architectural enhancements.
- **Developed a scalable ETL pipeline** for **AWS CloudWatch logs** to process IoT sensor metadata (connectivity, packet format, upload rate), leveraging Polars for high-performance transformation and **MongoDB joins** for enriched analytics integrated into **Apache Superset** dashboards.
- **Architected and deployed a Retrieval-Augmented Generation (RAG) system** using **OpenAI APIs** and **LlamaIndex**, designing a vector database schema, enabling multiprocessing-based embedding preloading, and delivering context-aware, structured query responses for Rekurv.
- **Automated CDP session management** using **Apache Airflow**, dynamically monitoring **Kafka partition lag** to auto-scale EC2 sessions and maintain performance, achieving a **30% reduction in infrastructure costs** and enhanced operational reliability.
- **Led the migration from Python 3.7 → 3.13**, refactoring modules to align with modern libraries and implementing a **multi-stage Docker build** that reduced image size from **1.54 GB to 800 MB**, while integrating **Astral's uv package manager** for faster builds and dependency resolution.
- **Enhanced predictive maintenance capabilities** by integrating ML models for vibration, ultrasonic, and temperature analysis, improving anomaly detection accuracy and enabling proactive fault diagnosis.
- **Optimized MongoDB interactions** by implementing pagination, projection, and bulk upsert strategies to reduce memory usage, prevent OOM issues, and enhance query efficiency across large datasets.
- **Refactored monolithic codebases into modular, scalable architectures using SOLID principles**, improving maintainability and extensibility, while driving system modernization by redesigning reset service logic across DTM/CDP systems and enabling seamless integration of advanced components such as Pattern Finder and NanoAI (FaultAmp) for next-generation fault detection.

Punctuations Solutions Pvt Ltd | Mumbai, India

March 2021 - September 2023

Software Engineer

- **Developed an Intelligent Document Processing (IDP) system** leveraging **OCR (Tesseract, PaddleOCR)** and **NLP (SpaCy)** to extract structured data from 2M+ documents, achieving a **4x improvement in processing efficiency** and storing extracted data in **Elasticsearch** for fast retrieval.
- **Implemented scalable microservices architecture** using **FastAPI, Celery, and RabbitMQ**, enabling parallel processing, asynchronous job handling, and high system availability across multiple nodes.
- **Built cloud storage connectors** for **Google Drive, Box, Dropbox, and Egnyte**, handling OAuth authentication, file/folder synchronization, and webhook-based real-time updates integrated with **Vue.js** frontend.
- **Designed a Question-and-Answer (Q&A) service** on top of Elasticsearch using **NLP pipelines** to interpret natural language queries and return precise form-based answers, enhancing client usability and automation.
- **Implemented a microservice-based object detection pipeline** using **YOLOv3/YOLOv4** and **TensorFlow**, improving borderless table and text block detection accuracy in scanned document images.
- **Developed an Identity Card extraction module** using **LayoutLM**, enhancing document understanding and improving **KYC verification accuracy** for enterprise clients.

- **Engineered WebSQL LLM platform** integrating with **Oracle, MySQL, PostgreSQL, and MariaDB**, automatically generating SQL queries from natural language, executing them, and visualizing results via interactive **Pie, Bar and Table charts**.
- **Integrated Redis caching and Elasticsearch indexing** to minimize query latency and enable near real-time dashboard updates, reducing query response time by **80%** while supporting dynamic visualization and analytics.

IT Dude Tech Solutions | Mumbai, India

April 2020 - February 2021

Python Developer

- **Developed a high-performance e-commerce platform using Django**, improving load times by **50%** through **query optimization and caching**, while implementing **secure authentication and payment processing** following industry best practices.
- **Designed and implemented a decentralized identity framework using Hyperledger Indy**, enabling **secure credential issuance, storage, and trusted exchange**, including an automated **passport data extraction and verification system** that improved accuracy by **30%**.
- **Engineered and optimized database architecture**, including **schema design, indexing, efficient query development, and backup strategies**, ensuring **data integrity, high availability, and performance at scale**.

PROJECT

REKURV – Retrieval-Augmented Generation (RAG)

- **Designed and developed a production-grade Retrieval-Augmented Generation (RAG) system**, enabling **accurate, context-aware responses** by combining enterprise data with LLM capabilities.
- **Architected the end-to-end pipeline**, including **vector database schema design, document ingestion, and embedding workflows**, and implemented **multiprocessing-based embedding preloading** to achieve **low-latency retrieval performance**.
- **Integrated LlamaIndex for efficient document chunking and semantic retrieval**, enabling dynamic fusion of **domain-specific data and general knowledge** for improved response relevance.
- **Leveraged OpenAI function calling** to generate **structured, schema-compliant outputs**, enhancing response accuracy, consistency, and downstream usability.

AIVISION – Intelligent Document Processing (IDP) System

- **Designed and developed an Intelligent Document Processing (IDP) platform** leveraging **OCR and NLP (SpaCy)** to extract, process, and analyze structured data from unstructured documents.
- **Architected a scalable microservices-based system** using **FastAPI, Celery, and RabbitMQ**, processing **2M+ documents efficiently**, with asynchronous background jobs and distributed task queues.
- **Built cloud storage connectors (Google Drive, Dropbox, Box)** enabling **secure authorization, file ingestion, and metadata extraction**, ensuring seamless integration across external data sources.
- **Developed a semantic Q&A system** powered by **NLP and Elasticsearch**, enabling **context-aware query resolution** from extracted document data.
- **Implemented Elasticsearch-based indexing and search optimization**, enabling **fast, scalable querying and analytics** on large volumes of processed documents.
- **Engineered a dynamic analytics dashboard using Vue.js**, delivering **customizable reports and real-time insights** tailored to client requirements.
- **Implemented JWT token authentication** to secure user access and integrated third-party APIs to enhance the application's functionality and data integration capabilities. **Designed independent OCR and object detection microservices**, enabling **modular deployment, horizontal scalability, and fault isolation**.
- **Implemented secure authentication using JWT tokens** and integrated **third-party APIs**, enhancing system security, interoperability, and feature extensibility.

EDUCATION

Mumbai University | Mumbai, India

CGPI - 8.27

Bachelor of Engineering in Computer Science

August 2016 - September 2020

CERTIFICATIONS

[Redis Python Developer](#)

[Apache Kafka Fundamentals](#)

[Generative AI Fundamentals](#)